

MDS220 Musculoskeletal System

HARD Practice Examination • Lectures 1 to 22 • Single Best Answer

87 high-difficulty questions: clinical vignettes, NOT/EXCEPT items, and nerve-muscle-action reasoning. All answer choices are written in full. Choose the single best answer.

SECTION A -- Histology, Cartilage, Bone, and Integument

1. A researcher fixes tissue in glutaraldehyde, stains with heavy metals, and wants to resolve the cristae of a single mitochondrion. Which instrument is required?
 - (A) Phase-contrast microscope
 - (B) Transmission electron microscope
 - (C) Confocal laser microscope
 - (D) Brightfield light microscope
 - (E) Scanning electron microscope
2. Haematoxylin and eosin staining separates tissue components based on which principle?
 - (A) Enzyme catalytic activity
 - (B) Carbohydrate periodate oxidation
 - (C) Metallic silver impregnation
 - (D) Cationic and anionic affinity
 - (E) Antibody antigen recognition
3. During routine paraffin processing, passing fixed tissue through graded ethanol accomplishes which step?
 - (A) Embedding
 - (B) Sectioning
 - (C) Dehydration
 - (D) Clearing
 - (E) Infiltration
4. In a toluidine-blue-stained neuron, the most intensely basophilic structure is the nucleolus because it is rich in?
 - (A) Ribosomal ribonucleic acid
 - (B) Membrane phospholipid
 - (C) Smooth reticulum lipid
 - (D) Golgi glycoprotein
 - (E) Mitochondrial enzyme
5. A skin punch biopsy from the reticular dermis is examined. Which tissue predominates?
 - (A) Loose areolar connective tissue
 - (B) Reticular connective tissue
 - (C) Elastic connective tissue
 - (D) Dense regular connective tissue
 - (E) Dense irregular connective tissue
6. In the tunica media of a large elastic artery, the elastin and collagen are synthesised by which cell?
 - (A) Smooth muscle cell
 - (B) Resident macrophage
 - (C) Endothelial cell
 - (D) Undifferentiated pericyte
 - (E) Vascular fibroblast
7. A child has repeated fractures, blue sclerae and progressive hearing loss. The defective collagen normally provides which property in its main tissues?
 - (A) Resistance to compression
 - (B) Basement membrane filtration
 - (C) Resistance to tension

- (D) Reticular scaffolding
- (E) Anchoring of the lamina

8. Which statement about connective tissue fibres is FALSE?

- (A) Reticular fibres are argyrophilic and periodic-acid positive
- (B) Elastic fibres provide greater tensile strength than collagen
- (C) Fibrillin microfibrils surround the elastin core
- (D) Type four collagen forms basement membrane meshwork
- (E) Collagen fibrils show sixty-eight nanometre banding

9. A tall patient with long limbs, lens dislocation and an aortic dissection has a defect in a protein that normally serves what role?

- (A) Cartilage matrix aggregate
- (B) Primary tension collagen
- (C) Reticular organ stroma
- (D) Scaffold for elastin deposition
- (E) Basement membrane mesh

10. Which glycosaminoglycan is non-sulfated and, being highly polyanionic, forms the backbone of the hydrated ground-substance gel?

- (A) Dermatan sulfate
- (B) Chondroitin sulfate
- (C) Heparan sulfate
- (D) Keratan sulfate
- (E) Hyaluronic acid

11. Which feature is UNIQUE to skeletal muscle among the three muscle types?

- (A) Ability to divide readily by mitosis
- (B) Triad at the anisotropic-isotropic junction
- (C) Presence of communicating gap junctions
- (D) Cross-striations of the filaments seen
- (E) Branching of the individual fibres

12. A structure that is NOT part of the neuromuscular junction is the?

- (A) Axon terminal
- (B) Synaptic cleft
- (C) Intrafusal fibre
- (D) Junctional fold
- (E) Motor end plate

13. An endurance marathoner's postural muscles are dominated by fibres with which property compared with type two-B fibres?

- (A) Higher force per twitch
- (B) Larger fibre diameter
- (C) Faster contraction speed
- (D) Higher oxidative capacity
- (E) Greater glycolytic capacity

14. In cardiac muscle the transverse tubule associates with a single sarcoplasmic cisterna at the Z-line to form a?

- (A) Diad
- (B) Dense body
- (C) Sarcomere
- (D) Caveola
- (E) Triad

15. Which characteristic is shared by BOTH skeletal and smooth muscle?

- (A) Cells are electrically coupled by gap junctions
- (B) Troponin protein regulates the thin filament

- (C) Spontaneous membrane depolarisation occurs
- (D) Contraction shows visible cross-striation
- (E) Rise in intracellular calcium triggers contraction

16. A patient develops fatigable weakness; antibodies block the postsynaptic receptor. The lesion is best localised to the?

- (A) Motor end plate
- (B) Sarcoplasmic reticulum
- (C) Ventral nerve root
- (D) Transverse tubule
- (E) Presynaptic terminal

17. Which organ's muscular wall contains NO smooth muscle at all?

- (A) Uterus
- (B) Urinary bladder
- (C) Stomach
- (D) Diaphragm
- (E) Muscular artery

18. Which statement about cartilage is TRUE?

- (A) Cartilage can grow by an intramembranous route
- (B) Fibrocartilage possesses a perichondrium
- (C) The hyaline territorial matrix is coarse
- (D) Fibrocartilage main collagen is type two
- (E) The matrix has both organic and inorganic components

19. Which feature is typical of elastic cartilage but NOT hyaline cartilage?

- (A) Complete absence of any perichondrium layer
- (B) Being the most widely distributed type
- (C) Abundant elastic fibres with collagen fibrils
- (D) Use as a template for future bone
- (E) Type one collagen predominance

20. Articular cartilage heals poorly after injury CHIEFLY because it?

- (A) Is richly vascular
- (B) Has no perichondrium
- (C) Lacks any chondrocytes
- (D) Contains elastic fibres
- (E) Grows only by apposition

21. Which statement about osteoblasts is FALSE?

- (A) They bear insulin-like growth factor receptors
- (B) They divide readily by mitosis
- (C) They differentiate later into resting osteocytes
- (D) They secrete the enzyme alkaline phosphatase
- (E) They actively form new bone matrix

22. In the epiphyseal growth plate, the zone in which chondrocytes undergo apoptosis and the matrix calcifies is the zone of?

- (A) Reserve
- (B) Proliferation
- (C) Calcification
- (D) Hypertrophy
- (E) Resting

23. The periosteal bone collar of a developing long bone forms?

- (A) During primary bone development
- (B) In the secondary ossification centre
- (C) Within the marrow cavity

- (D) During spongy bone growth
- (E) Only after epiphyseal fusion

24. A drug for osteoporosis that inhibits osteoclasts, but if used for many years blocks remodeling and predisposes to atypical fracture, works by making bone?

- (A) Soft and pliable
- (B) Low in mineral
- (C) Highly vascular
- (D) Hard but brittle
- (E) Rich in osteoid

25. The interstitial lamellae between intact osteons are best described as?

- (A) Newly formed circumferential bone
- (B) Zones of the growth plate
- (C) Volkmann canal linings
- (D) Uncalcified osteoid seams
- (E) Remnants of remodeled osteons

26. A hypodermic needle is inserted perpendicular to the skin. List the FIRST integumentary layer it pierces.

- (A) Hypodermis
- (B) Reticular dermis
- (C) Epidermis
- (D) Deep fascia
- (E) Papillary dermis

27. The stratum lucidum, present only in thick skin, lies immediately between which two layers?

- (A) Corneum and the surface
- (B) Granulosum and corneum
- (C) Basale and the dermis
- (D) Basale and spinosum
- (E) Spinosum and granulosum

28. A second-degree burn abolishes fine two-point touch yet spares the arrector pili motor response. The receptor most damaged is the?

- (A) Hair bulb
- (B) Pacinian corpuscle
- (C) Ruffini ending
- (D) Sebaceous gland
- (E) Meissner corpuscle

29. When core temperature rises, the autonomic response in the skin is to?

- (A) Constrict dermal vessels
- (B) Halt eccrine secretion
- (C) Shunt blood to deep veins
- (D) Dilate dermal vessels
- (E) Erect the pili muscles

30. Albinism is best explained by?

- (A) Inability to synthesise tyrosinase
- (B) Dietary lack of the amino acid tyrosine
- (C) Complete absence of all keratinocytes
- (D) Complete developmental loss of melanocytes
- (E) Overproduction of the pigment eumelanin

31. Which statement about a hair follicle is correct?

- (A) The follicle originates in the epidermis
- (B) The papilla is composed of dead cells
- (C) Hair originates in the subcutaneous fat

- (D) The shaft above skin is living tissue
- (E) Its bulb contains living dividing cells

SECTION B -- Gross Anatomy (Back, Upper and Lower Limb, Joints, Vertebral Column)

- 32.** A single muscle forms the inferior border of the triangle of auscultation and the medial border of the lumbar triangle. It is the?
- (A) Levator scapulae
 - (B) Latissimus dorsi
 - (C) Serratus posterior
 - (D) Splenius capitis
 - (E) Trapezius muscle
- 33.** A stab that severs the deep branch of the transverse cervical (dorsal scapular) supply causes ischaemic paralysis of the?
- (A) Longissimus muscle
 - (B) Trapezius muscle
 - (C) Latissimus dorsi
 - (D) Rhomboid major
 - (E) Semispinalis capitis
- 34.** The thoracodorsal nerve and artery supply which muscle?
- (A) Latissimus dorsi
 - (B) Serratus posterior
 - (C) Trapezius muscle
 - (D) Rhomboid major
 - (E) Levator scapulae
- 35.** The roof of the suboccipital triangle is formed by which muscle?
- (A) Splenius capitis
 - (B) Obliquus capitis inferior
 - (C) Semispinalis capitis
 - (D) Obliquus capitis superior
 - (E) Rectus capitis posterior major
- 36.** Which listed structure carries ONLY sensory fibres?
- (A) White ramus communicans
 - (B) Ventral root
 - (C) Dorsal root
 - (D) Spinal nerve
 - (E) Grey ramus communicans
- 37.** Which listed structure carries NO sensory fibres?
- (A) Ventral root
 - (B) Dorsal root
 - (C) White ramus communicans
 - (D) Dorsal ramus
 - (E) Spinal nerve
- 38.** A fracture through the pars interarticularis without vertebral slippage is termed?
- (A) Spondylitis
 - (B) Spondylolysis
 - (C) Spondylolisthesis
 - (D) Spondylosis
 - (E) Spina bifida
- 39.** A rotator-cuff muscle that both initiates abduction and is supplied by the suprascapular nerve is the?

- (A) Subscapularis
- (B) Infraspinatus
- (C) Deltoid muscle
- (D) Teres minor
- (E) Supraspinatus

40. A lesion in the quadrangular space would MOST directly affect which nerve and its motor territory?

- (A) Long thoracic nerve and serratus
- (B) Musculocutaneous nerve and biceps
- (C) Radial nerve and triceps
- (D) Axillary nerve and deltoid
- (E) Suprascapular nerve and cuff

41. A modified radical mastectomy paralyses the lower sternocostal fibres of pectoralis major. The nerve injured is the?

- (A) Lateral pectoral nerve
- (B) Long thoracic nerve
- (C) Medial pectoral nerve
- (D) Thoracodorsal nerve
- (E) Lower subscapular nerve

42. During axillary surgery the long thoracic nerve is cut. The resulting deformity is a?

- (A) Trendelenburg gait
- (B) Wrist drop
- (C) Claw hand
- (D) Ape hand
- (E) Winged scapula

43. A patient fractures the surgical neck of the humerus. The area of cutaneous anaesthesia most expected is the?

- (A) Medial side of the arm
- (B) Lateral side of the arm
- (C) Medial side of forearm
- (D) Dorsum of the hand
- (E) Lateral side of forearm

44. A mid-shaft (distal third) humeral fracture from a motorcycle crash most likely requires checking which function?

- (A) Elbow flexion
- (B) Biceps contraction
- (C) Finger flexion
- (D) Wrist extension
- (E) Thumb adduction

45. Passing a needle at the apex of the cubital fossa, just medial to the biceps tendon, which structure is LEAST likely to be encountered?

- (A) Brachioradialis muscle
- (B) The bicipital aponeurosis
- (C) The brachial artery
- (D) The median nerve trunk
- (E) The median cubital vein

46. Which muscle-and-nerve pairing is CORRECT?

- (A) Flexor digitorum superficialis with anterior interosseous nerve
- (B) Extensor carpi radialis brevis with median nerve
- (C) Flexor pollicis longus with the deep radial nerve
- (D) Extensor indicis with posterior interosseous nerve
- (E) Pronator teres paired with the ulnar nerve

47. A laceration lateral to the palmaris longus tendon at the wrist that abolishes thumb opposition and lateral palmar sensation has divided the?

- (A) Radial nerve
- (B) Ulnar nerve
- (C) Anterior interosseous nerve
- (D) Superficial radial nerve
- (E) The median nerve trunk

48. A fall fractures the scaphoid. Beyond snuffbox tenderness, the feared complication is avascular necrosis because the scaphoid's blood supply enters?

- (A) Proximally, leaving the distal pole at risk
- (B) Distally, leaving the proximal pole at risk
- (C) Through the palmar surface only
- (D) From the ulnar artery alone
- (E) Via the deep palmar arch only

49. The anatomical snuffbox contains which structure on its floor?

- (A) Second part of the radial artery
- (B) First part of the ulnar artery
- (C) The recurrent branch of median nerve
- (D) Deep branch of the ulnar nerve
- (E) The superficial palmar arterial arch

50. A fall onto the point of the shoulder separates the acromioclavicular joint. The ligament whose tearing most determines instability is the?

- (A) Costoclavicular ligament
- (B) Transverse humeral ligament
- (C) Coracoclavicular ligament
- (D) Coracoacromial ligament
- (E) Acromioclavicular ligament

51. In a child, a sudden pull on the pronated forearm subluxes the radial head from which ligament?

- (A) Radial collateral ligament
- (B) Oblique cord
- (C) Ulnar collateral ligament
- (D) Interosseous membrane
- (E) Annular ligament

52. When a patient stands on the right leg the left (unsupported) pelvis drops. The lesion is in the nerve supplying which muscle group?

- (A) Left hip adductors
- (B) Right hip abductors
- (C) Right hip extensors
- (D) Right knee extensors
- (E) Left hip abductors

53. Which listed muscle does NOT laterally rotate the hip joint?

- (A) Superior gemellus
- (B) Inferior gemellus
- (C) Pectineus muscle
- (D) Piriformis muscle
- (E) Quadratus femoris

54. A fracture of the anterior inferior iliac spine that impairs hip flexion and knee extension paralyses which muscle?

- (A) Vastus lateralis
- (B) Sartorius muscle
- (C) Iliacus muscle

- (D) Rectus femoris
- (E) Tensor fasciae latae

55. Compression of the obturator nerve in the obturator canal will NOT produce which finding?

- (A) Reduced knee-jerk reflex
- (B) Loss of medial thigh sensation
- (C) Weakness of thigh adduction
- (D) Referred groin pain
- (E) Weakness of adductor brevis

56. The adductor (subsartorial) canal transmits the femoral vessels, the nerve to vastus medialis, and which additional nerve?

- (A) Sciatic nerve
- (B) Obturator nerve
- (C) Sural nerve
- (D) Tibial nerve
- (E) Saphenous nerve

57. The medial compartment of the femoral sheath, the femoral canal, normally contains?

- (A) The great saphenous venous trunk
- (B) The obturator nerve branch
- (C) The femoral artery trunk
- (D) Loose tissue and lymphatics
- (E) The femoral nerve trunk

58. All hamstrings share a tibial-nerve supply EXCEPT one head, which is supplied by the common fibular division. That head is the?

- (A) Short head of biceps femoris
- (B) Semitendinosus muscle
- (C) Adductor magnus hamstring part
- (D) Long head of biceps femoris
- (E) Semimembranosus muscle

59. A penetrating wound to the popliteal fossa most endangers which most superficial major structure?

- (A) Common fibular nerve
- (B) Tibial nerve
- (C) Small saphenous vein
- (D) Popliteal vein
- (E) Popliteal artery

60. The boundary of the popliteal fossa formed by the medial head of gastrocnemius is the?

- (A) Superomedial border
- (B) Inferolateral border
- (C) Inferomedial border
- (D) Superolateral border
- (E) Roof of the fossa

61. A patient cannot invert the foot after an injury. Loss of BOTH the tibialis anterior and tibialis posterior implicates which two nerves?

- (A) Sural and saphenous nerves
- (B) Deep and superficial fibular nerves
- (C) Superficial fibular and tibial nerves
- (D) Medial and lateral plantar nerves
- (E) Deep fibular and tibial nerves

62. A construction worker fractures the groove beneath the sustentaculum tali. Which tendon is most likely torn?

- (A) Tibialis posterior
- (B) Fibularis longus

- (C) Flexor digitorum longus
- (D) Flexor hallucis longus
- (E) Flexor hallucis brevis

63. Thickening of the posterior surface of the distal two-thirds of the tibia (the soleal line region) most affects which pair of muscles?

- (A) Tibialis posterior and flexor hallucis longus
- (B) Soleus and tibialis posterior
- (C) Gastrocnemius and tibialis posterior
- (D) Soleus and flexor digitorum longus
- (E) Gastrocnemius and the soleus

64. The transverse arch of the foot is maintained by the tibialis posterior and which other tendon?

- (A) Fibularis longus
- (B) Fibularis brevis
- (C) Flexor digitorum brevis
- (D) Extensor hallucis longus
- (E) Tibialis anterior

65. Which structure is NOT contained in the central compartment of the sole?

- (A) Adductor hallucis
- (B) The lumbrical muscles
- (C) Medial plantar artery
- (D) Quadratus plantae
- (E) Flexor digitorum brevis

66. A deep fibular nerve palsy is confirmed clinically by inability to?

- (A) Evert the foot
- (B) Dorsiflex the foot
- (C) Invert the foot
- (D) Flex the toes
- (E) Plantarflex the foot

67. A dashboard injury drives the femur posteriorly; which hip ligament, resisting that displacement, is torn?

- (A) Transverse acetabular ligament
- (B) Iliofemoral ligament
- (C) Ligament of the head
- (D) Ischiofemoral ligament
- (E) Pubofemoral ligament

68. A skier catches an edge and the tibia is driven forward on the femur during sudden deceleration. Which ligament most likely ruptures?

- (A) Posterior cruciate ligament
- (B) Fibular collateral ligament
- (C) Anterior cruciate ligament
- (D) Tibial collateral ligament
- (E) Patellar ligament

69. A blow to the lateral knee tears the tibial collateral ligament. Which structure is also injured owing to its attachment?

- (A) Anterior cruciate ligament
- (B) Posterior cruciate ligament
- (C) Popliteus tendon
- (D) Lateral meniscus
- (E) Medial meniscus

70. Infection in which bursa is LEAST likely to spread into the knee joint cavity?

- (A) Semimembranosus bursa

- (B) Deep infrapatellar bursa
- (C) Gastrocnemius bursa
- (D) Suprapatellar bursa
- (E) Popliteus bursa

71. A fractured neck of the talus most likely injures which ligament?

- (A) Calcaneofibular ligament
- (B) Anterior talofibular ligament
- (C) Posterior tibiofibular ligament
- (D) Long plantar ligament
- (E) Deltoid ligament

72. The knee is unlocked from full extension by lateral rotation of the femur on the fixed tibia produced by which muscle?

- (A) Quadriceps femoris
- (B) Biceps femoris
- (C) Popliteus muscle
- (D) Sartorius muscle
- (E) Gastrocnemius muscle

73. A whiplash (severe hyperextension) injury most often tears which ligament?

- (A) The ligamentum flavum
- (B) The supraspinous ligament
- (C) Posterior longitudinal ligament
- (D) Interspinous ligament
- (E) Anterior longitudinal ligament

74. In cervical disc herniation without cord compression, the ligament shielding the cord is the?

- (A) The supraspinous ligament
- (B) Anterior longitudinal ligament
- (C) The ligamentum flavum
- (D) Posterior longitudinal ligament
- (E) The nuchal ligament

75. Spinal stenosis in an older patient most often results from hypertrophy of which structure?

- (A) The ligamentum flavum
- (B) The nuchal ligament
- (C) The supraspinous ligament
- (D) Anterior longitudinal ligament
- (E) Posterior longitudinal ligament

76. A collision fractures the dens of the axis. Normally the dens is held against the anterior arch of the atlas by the?

- (A) The ligamentum flavum
- (B) Transverse ligament of atlas
- (C) Alar ligament
- (D) Anterior longitudinal ligament
- (E) Tectorial membrane

77. Ninety-five percent of lumbar disc herniations occur at L4-L5 or L5-S1 and compress which nerve component to cause sciatica?

- (A) The twelfth thoracic nerve root
- (B) The fourth sacral nerve root
- (C) The first lumbar nerve root
- (D) The third and fourth lumbar roots
- (E) The fifth lumbar or first sacral root

78. There is no intervertebral disc between which two vertebrae?

- (A) The fifth lumbar and sacrum

- (B) The first and second thoracic
- (C) The atlas and the axis
- (D) The seventh cervical and first thoracic
- (E) The axis and the third cervical

SECTION C -- Embryology and Development

- 79.** A newborn has a long, narrow, wedge-shaped skull. The prematurely fused suture is the?
- (A) Lambdoid suture
 - (B) Squamous suture
 - (C) Coronal suture
 - (D) Sagittal suture
 - (E) Metopic suture
- 80.** An infant has a high, tower-like skull from bilateral premature closure of which suture?
- (A) Coronal suture
 - (B) Metopic suture
 - (C) Sagittal suture
 - (D) Squamous suture
 - (E) Lambdoid suture
- 81.** The malleus and incus arise from the cartilage of the first pharyngeal arch, which is?
- (A) The otic capsule
 - (B) Reichert cartilage
 - (C) Parachordal cartilage
 - (D) The trabecular cartilage
 - (E) Meckel cartilage
- 82.** Which cranial bone forms by intramembranous ossification within the membranous neurocranium?
- (A) Frontal bone
 - (B) Basiocciput
 - (C) Petrous temporal
 - (D) Ethmoid bone
 - (E) Sphenoid bone
- 83.** Failure of interdigital web apoptosis produces which digital anomaly?
- (A) Brachydactyly
 - (B) Clubfoot
 - (C) Syndactyly
 - (D) Polydactyly
 - (E) Cleft hand
- 84.** A neonate with a lobster-claw hand missing one or more central digits has which defect?
- (A) Amelia
 - (B) Cleft hand
 - (C) Polydactyly
 - (D) Brachydactyly
 - (E) Syndactyly
- 85.** The dermis over the back is derived from which mesoderm, in contrast to the limb dermis?
- (A) Intermediate mesoderm
 - (B) Lateral plate mesoderm
 - (C) Splanchnic mesoderm
 - (D) Paraxial mesoderm
 - (E) Neural crest cells
- 86.** A muscle NOT derived from the hypaxial division of the myotome is the?
- (A) The infrahyoid strap musculature
 - (B) Rectus abdominis muscle

- (C) External oblique muscle
- (D) Transversospinalis muscle
- (E) The appendicular limb musculature

87. The nucleus pulposus and hemivertebra relate to which pair of embryonic structures, respectively?

- (A) Dermatome and the notochord
- (B) Neural crest and the arch
- (C) Notochord and the centrum
- (D) Sclerotome and the myotome
- (E) Myotome and the centrum

Answer Key with Explanations

Keys derived programmatically from concept anchors and verified.

SECTION A -- Histology, Cartilage, Bone, and Integument

- 1. B — Transmission electron microscope**
Only the transmission electron microscope resolves internal organelle ultrastructure such as mitochondrial cristae; the scanning type shows surface only.
- 2. D — Cationic and anionic affinity**
Haematoxylin (basic, cationic) binds acidic/basophilic structures; eosin (acidic, anionic) binds basic/acidophilic ones. It is acid-base chemistry.
- 3. C — Dehydration**
Graded ethanol removes water (dehydration) before clearing in a solvent and embedding in paraffin.
- 4. A — Ribosomal ribonucleic acid**
The nucleolus is dense in ribosomal RNA whose phosphate groups are strongly acidic, so it binds basic dyes most avidly.
- 5. E — Dense irregular connective tissue**
The reticular dermis is dense irregular connective tissue with multidirectional thick collagen; the papillary dermis above it is loose.
- 6. A — Smooth muscle cell**
In large vessel walls the smooth muscle cells, not fibroblasts, produce the elastin and collagen of the media.
- 7. C — Resistance to tension**
This is osteogenesis imperfecta (type I collagen). Type I collagen resists tension in bone, tendon and dermis.
- 8. B — Elastic fibres provide greater tensile strength than collagen**
Elastic fibres have LESS tensile strength than collagen but recoil after stretch; the other statements are correct.
- 9. D — Scaffold for elastin deposition**
Marfan syndrome is a fibrillin-1 defect; fibrillin microfibrils form the scaffold on which elastin is deposited.
- 10. E — Hyaluronic acid**
Hyaluronic acid is the large non-sulfated glycosaminoglycan onto which proteoglycans aggregate, holding water as a gel.
- 11. B — Triad at the anisotropic-isotropic junction**
Triads at the A-I junction are unique to skeletal muscle; striations are shared with cardiac, and branching/gap junctions belong to cardiac.
- 12. C — Intrafusal fibre**
Intrafusal fibres belong to the muscle spindle (a sensory proprioceptor), not the motor neuromuscular junction.
- 13. D — Higher oxidative capacity**
Type I (slow) fibres have more mitochondria, myoglobin and capillaries, giving higher oxidative capacity for endurance.
- 14. A — Diad**
Cardiac muscle has diads (one T-tubule plus one cisterna) at the Z-line; skeletal muscle has triads at the A-I junction.
- 15. E — Rise in intracellular calcium triggers contraction**
All muscle contracts when intracellular calcium rises. Troponin/striation are skeletal-specific; gap junctions and autorhythmicity are smooth/cardiac.
- 16. A — Motor end plate**
Myasthenia gravis antibodies target acetylcholine receptors on the postsynaptic motor end plate, weakening transmission.
- 17. D — Diaphragm**
The diaphragm is entirely skeletal muscle; the bladder, arteries, uterus and stomach all contain smooth muscle.
- 18. E — The matrix has both organic and inorganic components**
Cartilage matrix has organic fibres/ground substance and some inorganic content. Fibrocartilage lacks perichondrium and uses type I collagen.
- 19. C — Abundant elastic fibres with collagen fibrils**
Elastic cartilage adds elastic fibres to its type II collagen; it retains a perichondrium and is not a bone template.

20. B — Has no perichondrium

Articular cartilage lacks a perichondrium and is avascular, so the chondrogenic source and blood supply for repair are absent.

21. B — They divide readily by mitosis

Mature osteoblasts do not divide; they arise from mitotic osteoprogenitor cells and mature into osteocytes.

22. C — Calcification

Beyond hypertrophy, the zone of calcification is where chondrocytes die and the matrix calcifies before ossification.

23. A — During primary bone development

Osteoprogenitor cells of the perichondrium form the periosteal collar during primary (diaphyseal) endochondral ossification.

24. D — Hard but brittle

Prolonged bisphosphonate use blocks osteoclast-driven remodeling, leaving bone hard but brittle and prone to atypical fracture.

25. E — Remnants of remodeled osteons

Interstitial lamellae are leftover fragments of previous osteons partially removed during remodeling.

26. C — Epidermis

From the surface inward the needle passes epidermis first, then papillary and reticular dermis, then hypodermis.

27. B — Granulosum and corneum

In thick (glabrous) skin the clear stratum lucidum sits between the stratum granulosum below and the stratum corneum above.

28. E — Meissner corpuscle

Meissner corpuscles in the superficial dermal papillae mediate fine touch and are lost in partial-thickness burns.

29. D — Dilate dermal vessels

Heat causes dermal vasodilation and sweating for heat loss; arrector pili relax and shivering stops.

30. A — Inability to synthesise tyrosinase

In albinism melanocytes are present but cannot make tyrosinase, so melanin synthesis fails.

31. E — Its bulb contains living dividing cells

Only the hair bulb contains living, dividing matrix cells; the follicle originates from dermis and the shaft is keratinised dead tissue.

SECTION B -- Gross Anatomy (Back, Upper and Lower Limb, Joints, Vertebral Column)

32. B — Latissimus dorsi

Latissimus dorsi contributes to both the triangle of auscultation and the lumbar (Petit) triangle boundaries.

33. D — Rhomboid major

The deep branch of the transverse cervical artery runs with the dorsal scapular nerve to the rhomboids.

34. A — Latissimus dorsi

The thoracodorsal nerve and artery both supply latissimus dorsi.

35. C — Semispinalis capitis

Semispinalis capitis (with splenius) forms the roof; the floor is the posterior atlanto-occipital membrane and posterior arch of the atlas.

36. C — Dorsal root

The dorsal root carries only afferent sensory fibres; the ventral root is motor and the spinal nerve is mixed.

37. A — Ventral root

The ventral root is purely motor (somatic and preganglionic autonomic efferents); it carries no sensory fibres.

38. B — Spondylolysis

Spondylolysis is a pars interarticularis defect; if the vertebra then slips it becomes spondylolisthesis.

39. E — Supraspinatus

Supraspinatus initiates the first fifteen degrees of abduction and is supplied by the suprascapular nerve.

40. D — Axillary nerve and deltoid

The quadrangular space transmits the axillary nerve (to deltoid and teres minor) and the posterior circumflex humeral artery.

41. C — Medial pectoral nerve

The medial pectoral nerve supplies pectoralis minor and the lower part of pectoralis major, so its loss weakens the lower fibres.

42. E — Winged scapula

Long thoracic nerve injury paralyses serratus anterior, producing a winged scapula.

43. B — Lateral side of the arm

The axillary nerve, endangered at the surgical neck, gives the upper lateral cutaneous nerve of the arm over the deltoid (regimental badge).

44. D — Wrist extension

The radial nerve lies in the spiral groove at the humeral mid-shaft; its injury causes wrist drop, so wrist extension is checked.

45. A — Brachioradialis muscle

Brachioradialis forms the lateral boundary, so it is not found medial to the biceps tendon; the artery, median nerve and aponeurosis are.

46. D — Extensor indicis with posterior interosseous nerve

Extensor indicis is a posterior interosseous (deep radial) muscle; the other pairings are wrong.

47. E — The median nerve trunk

The median nerve lies just lateral to palmaris longus; its division causes an ape hand (loss of opposition) and lateral palmar numbness.

48. B — Distally, leaving the proximal pole at risk

The scaphoid is supplied distal-to-proximal, so a waist fracture can devascularise the proximal pole, causing avascular necrosis.

49. A — Second part of the radial artery

The second part of the radial artery crosses the snuffbox floor (scaphoid and trapezium).

50. C — Coracoclavicular ligament

The coracoclavicular ligament (conoid and trapezoid) is the key vertical stabiliser; its rupture gives significant separation.

51. E — Annular ligament

Pulled elbow (nursemaid's elbow) slips the radial head out of the annular ligament of the proximal radioulnar joint.

52. B — Right hip abductors

A drop of the unsupported side indicates weak abductors on the WEIGHT-BEARING (right) side, a right superior gluteal nerve lesion (Trendelenburg).

53. C — Pectineus muscle

Pectineus is chiefly a hip flexor and adductor; the gemelli, piriformis and quadratus femoris are lateral rotators.

54. D — Rectus femoris

Rectus femoris arises from the anterior inferior iliac spine and crosses two joints, flexing the hip and extending the knee.

55. A — Reduced knee-jerk reflex

The knee jerk is femoral (quadriceps, L2-L4); the obturator nerve injury spares it but weakens adduction and medial-thigh sensation.

56. E — Saphenous nerve

The saphenous nerve travels in the adductor canal and emerges to supply the medial leg and foot.

57. D — Loose tissue and lymphatics

The femoral canal holds fat, loose connective tissue and lymphatics, permitting venous distension; a femoral hernia enters here.

58. A — Short head of biceps femoris

The short head of biceps femoris is uniquely common-fibular; the rest of the hamstrings are tibial.

59. B — Tibial nerve

In the popliteal fossa the tibial nerve is most superficial, then the vein, with the artery deepest against the femur.

60. C — Inferomedial border

The medial head of gastrocnemius forms the inferomedial border; the lateral head forms the inferolateral border.

61. E — Deep fibular and tibial nerves

Inversion uses tibialis anterior (deep fibular) and tibialis posterior (tibial), so both nerves must be affected.

62. D — Flexor hallucis longus

The flexor hallucis longus tendon runs in the groove beneath the sustentaculum tali.

63. B — Soleus and tibialis posterior

The soleal line and posterior tibia give origin to soleus and tibialis posterior.

64. A — Fibularis longus

The fibularis longus and tibialis posterior tendons cross the sole to maintain the transverse arch.

65. C — Medial plantar artery

The medial plantar vessels lie in the medial compartment; the central compartment holds the flexor digitorum brevis, quadratus plantae, lumbricals and adductor hallucis.

66. B — Dorsiflex the foot

The deep fibular nerve supplies the anterior compartment dorsiflexors, so its palsy causes foot drop with lost dorsiflexion.

67. D — Ischiofemoral ligament

The posterior ischiofemoral ligament is torn in posterior hip dislocation from a dashboard force.

68. C — Anterior cruciate ligament

Sudden deceleration with anterior tibial translation typically ruptures the anterior cruciate ligament.

69. E — Medial meniscus

The tibial (medial) collateral ligament is attached to the medial meniscus, so both are injured together.

70. A — Semimembranosus bursa

The semimembranosus bursa usually does not communicate with the knee cavity, unlike the suprapatellar, popliteus and gastrocnemius bursae.

71. B — Anterior talofibular ligament

The anterior talofibular ligament is closely related to the talar neck and is commonly injured in such fractures.

72. C — Popliteus muscle

Popliteus laterally rotates the femur on the tibia (or medially rotates the tibia) to unlock the extended knee.

73. E — Anterior longitudinal ligament

The anterior longitudinal ligament resists hyperextension and is stretched or torn in whiplash.

74. D — Posterior longitudinal ligament

The posterior longitudinal ligament reinforces the disc centrally and helps prevent posterior herniation onto the cord.

75. A — The ligamentum flavum

Hypertrophy of the elastic ligamentum flavum narrows the canal, a common cause of acquired spinal stenosis.

76. B — Transverse ligament of atlas

The transverse ligament of the atlas retains the dens; the alar ligaments limit rotation and the tectorial membrane is a posterior longitudinal continuation.

77. E — The fifth lumbar or first sacral root

Posterolateral herniation at these levels typically compresses the L5 or S1 root, producing sciatica.

78. C — The atlas and the axis

There is no intervertebral disc between C1 (atlas) and C2 (axis).

SECTION C -- Embryology and Development

79. D — Sagittal suture

Scaphocephaly results from premature sagittal suture fusion, restricting width and forcing anteroposterior growth.

80. A — Coronal suture

Bilateral coronal synostosis produces brachycephaly, a short broad tower-like skull.

81. E — Meckel cartilage

Meckel cartilage of the first arch forms the malleus and incus; Reichert cartilage of the second forms the stapes and styloid process.

82. A — Frontal bone

The frontal and parietal bones and the interparietal occiput form intramembranously (membranous neurocranium).

83. C — Syndactyly

Persistence of the interdigital webs from failed apoptosis leaves fused digits (syndactyly).

84. B — Cleft hand

Cleft (lobster-claw) hand or foot is absence of one or more central digits.

85. D — Paraxial mesoderm

Back (dorsal) dermis comes from paraxial mesoderm (dermatome); limb and body-wall dermis from lateral plate mesoderm.

86. D — Transversospinalis muscle

Epaxial derivatives such as transversospinalis come from the epaxial division (dorsal rami); the rest are hypaxial (ventral rami).

87. C — Notochord and the centrum

The nucleus pulposus derives from the notochord; hemivertebra arises from a chondrification failure of the centrum.